

Hyuk Che (Luke) Kwon

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Education

University of Pennsylvania Philadelphia, PA	Aug. 2025 - May 2027
Master of Science in Engineering, Computer Graphics and Game Technology	GPA: 3.90
Sogang University Seoul, South Korea	Mar. 2018 - Aug. 2024
Bachelor of Science in Engineering, Computer Science and Engineering	Major GPA: 3.92
Bachelor of Arts, Global Korean Studies	Magna Cum Laude

Programming Skills

Languages: C, C++, GLSL, Python, F#

Tools/Frameworks: OpenGL, Nvidia Nsight, RenderDoc, GLM, Maya, Blender, Houdini, Git, Visual Studio, Qt Creator

Projects

Deferred Renderer with Screen Space Reflections | C++, OpenGL, GLSL, Nvidia Nsight Apr. 2026

- Built a deferred rendering pipeline storing surface attributes to a multi-attachment G-buffer, achieving a 4.4× speedup over forward rendering on scenes with many lights and heavy overdraw.
- Implemented SSR via G-buffer ray marching, with multi-level Gaussian blur for varying surface roughness.
- Precomputed diffuse and split-sum specular IBL convolutions, yielding a 5.1× speedup at higher sample quality.

Monte Carlo Path Tracer | C++, OpenGL, GLSL Feb. 2026

- Developed a Monte Carlo path tracer solving the Light Transport Equation for diffuse, specular, and dielectric materials.
- Built a Multiple Importance Sampling integrator using the Power Heuristic to combine direct BSDF and light sampling.
- Implemented a full global illumination integrator computing MIS direct lighting at every bounce with BSDF-sampled indirect lighting and environment map support.

Space Voxel Game | C++, OpenGL, GLSL, RenderDoc Dec. 2025

- Collaborated with two teammates to create a spherical minecraft world in C++, using OpenGL.
- Implemented player physics by incorporating raymarching to support collision detection on a curved environment.
- Sampled 3D Perlin noise for cave systems, and built a post-processing pipeline for underwater effects.
- Created a procedural skybox using 3D Worley noise to generate stars and 3D Perlin noise-based FBM for nebulae.

CPU Rasterizer | C++, GLM Sep. 2025

- Developed a 3D graphics engine in C++ featuring a custom perspective camera, rasterization, and Z-buffering for depth.
- Utilized perspective-correct barycentric interpolation for accurate color, texture mapping, and depth.
- Implemented Lambertian and Toon shading, supersampling anti-aliasing, and Bresenham's line algorithm.

Experience

Teaching Assistant | *Introduction to Computer Graphics, University of Pennsylvania* Jan. 2026 - Present

- Assisted in teaching graphics programming topics including matrix transformations, image processing, and rasterization.
- Held office hours to answer conceptual questions and assist assignment debugging in C++.

Undergraduate Research Assistant | *Visual Computing Lab, Sogang University* Apr. - June 2024

- Coordinated with a supervising professor to deliver weekly research presentations on view-synthesis and feature matching papers including SIFT, NeRF, 3D Gaussian Splatting, and Scaffold-GS.

Volunteering

ACM SIGGRAPH Student Volunteer 2025 Vancouver, 2024 Tokyo

Aug. 2025, Dec. 2024